

Final Exam Practice

Regression Analysis Scenario: Predicting House Prices

Issues with the model:

- Predictors are not standardized. This means that we can't directly assess the influence of the variables by looking at β s alone, since the β s are in terms of different units.
- Multicollinearity between predictors - Prediction that as Square Feet and Price are likely to have high covariance, he should account for this covariance via an interaction term: Square Feet $\times$ Price.
- Outliers: House 15 has a Square Feet much greater than other houses, and it also has the greatest Distance Downtown and is priced far above the rest of the houses. This outlier may skew the model coefficients, and inflating the significance of Square Feet and Distance Downtown as predictors of price.
- Assumptions - The OLS model is only accurate on linear data, he should verify that the data is indeed linear before proceeding with the model. If the data is not linear, perhaps he should use a different regression model, such as logistic regression.

Featureization and Data Representation

Explaining:

- Square Feet
- Bedrooms
- Distance to Water

Response: Price

Part 1.

1. Nominal: Age, income, avg. daily, commute - min, transactions - ln, remote - work days - per

Categorical: Neighborhood, owns-car, transit-card, preferred - transportation

ordinal: None

Identifiers: Comm-id

Free-target: notes

2. • Neighborhood, transit-card, owns-car have missing values
• Notes may have to be created -> a categorical variable, or dropped entirely
3. • C005 has None for transit-card
• C009 has a missing value for owns-car
• C006 has a missing value for neighborhood
4. Notes - high correlation between responses
comm-id - spurious trends can be found from identifier

Part 2.

1. No, as Gender-id is simply an identifier associated with each person in the dataset. If we have to include it in our model, the model may find spurious associations between Gender-id and the response, when in fact no such association actually exists in the population.
2. One-hot encoding: Assign each kind of neighborhood a column (e.g. Northside, West End, etc.). Then, each row will have 1 under that neighborhood's column if the Gender belongs to that neighborhood.

Ex.

Gender-id	...	Northside	West End	Downtown	...
C001		1	0	0	
C002		0	1	0	
C003	...	0	0	1	...
⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮

Label encoding: Assign each kind of neighborhood a label.
(e.g. Northside = 1, West End = 2, etc.)

- One-hot encoding - leads to a significant increase in the number of predictors, as each neighborhood now populates a column, which can lead to overfitting.
- Label encoding - assigns an ordering to neighborhoods by assigning each neighborhood to a label, but in fact no such ordering exists. This can make interpreting the model's results difficult.

- 3.
- Preferred transport is categorical - need to decide it via one-hot encoding or label encoding
 - Mixed category is ambiguous - does not specify exactly which combination of transport is preferred.
 - If mixed is significant, we can only say that many commuters prefer multiple modes of transport

- 4.
- No, notes should not be used in an imputation model, as there is high variation in responses
 - We could run a separate classifier that predicts whether each note suggests the commuter is of high income or not, but this introduces an additional layer of complexity in the model, and makes the model less generalizable
 - The risks of using free-text data are
 1. High variation in responses
 2. Responses are not always indicative of what we want to predict
 3. Including free-text text in our model adds an additional layer of complexity, making our model less generalizable

5. 1. There may be an influential feature that skews the income predictor.

2.

- Normalization - Even with normalization, this point still can be an influential point
- Regression models - Can skew model predictions to heavily weight predictions by income, when such an assumption actually does not

exam in the population.

- Distance-based models such as K-nearest also can put too much weight on high income due to this point. For example, the centroid can get pulled outside to higher income thresholds due to this point.

- 3.
- Control income to purchase (decrease efficiency, increase robustness)
 - Regularization (penalize large model weights)
 - Collect more data to decrease the outlier's leverage on the model

Part 3.

- Non-response - perhaps the genders are unrelated to people taking fields, or they did not want to fill out the missing fields
- The missing values should not be removed, as then the whole data point would have to be omitted from the dataset; this is costly since the sample size is already small.

Treating missing data as "unknown" is probably the best strategy as the data point can not only be kept, but also expectations can perhaps be observed to learn the response variable and those missing values

3. Yes. It is very possible that fields with missing responses follow some sort of pattern (ex. those who don't have a car may be more likely to skip the survey questions than to answer "no").

Evaluating a Classification Pipeline for Fraud Detection

Issues with the pipeline

1. True join between transactions and customer-profiles

requires any customer-ids that appear in one

dataset but not the other

- It may be possible that a fraudulent transaction has done without a customer profile, and the the model can use this record to better predict fraudulent transactions

2. Dataset imbalance between fraudulent and non-fraudulent transactions

- Since non-fraudulent transactions far outweigh fraudulent transactions even models that predict "not fraudulent" on every transaction can achieve a very high accuracy.
 - Can weigh a fraudulent transaction more than non-fraudulent transaction
 - Can use recall of the fraudulent class to better see how the model is identifying fraudulent transactions specifically

3. Possible collinearity in amount and num-items as predictors

- Can use interaction term amount: num-items to capture high covariance in predictors

PCA

- Online users has the least standardized variance, so it should have the least contribution to the PC
- Age, Amazon and Spotify score are all positively correlated, so we would expect their PC components to all follow the same sign.

So, a reasonable PL_1 would be:

$$PL_1 = \begin{bmatrix} \text{Age} \\ \text{Annual Income} \\ \text{Spending Score} \\ \text{Lifetime Value} \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.5 \\ 0.2 \\ 0.1 \end{bmatrix}$$